

# BiteFixes Enterprise Architecture Manual

## Volume 06

### Workflow Engine Specification (WFES)

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## Chapter 1

### Introduction

The Workflow Engine is responsible for executing business processes.

Business logic must never be hardcoded inside controllers.

Instead, workflows describe the behavior of the platform.

# Chapter 2

## Workflow Philosophy

A workflow is a business process.

Not a Python function.

Not an API endpoint.

Not a SQL procedure.

It is a sequence of business states.

Business Event



Workflow



States



Transitions



Actions



Events



Completion

## Chapter 3

# Workflow Components

Every workflow contains:

Workflow

↓

Metadata

↓

States

↓

Transitions

↓

Conditions

↓

Actions

↓

Timers

↓

Escalations



Permissions



Audit

## Chapter 4

# State Machine

Every workflow is a finite state machine.

Example

Repair Ticket

Draft



Open



Diagnosis



Approval



Waiting Parts



Repair



Quality Check



Ready



Delivered



Closed

Only valid transitions are allowed.

## Chapter 5

# Workflow Definition

Every workflow contains:

Workflow ID

Name

Description

Domain

Version

Status

Initial State

Final States

Timeout Policy

Owner

Created Date

## Chapter 6

### State Definition

Every state defines:

State ID

State Name

Description

Entry Actions

Exit Actions

Allowed Roles

Timeout

SLA

Notifications

Example

State

Diagnosis

Entry

Assign Technician

Exit

Generate Estimate

Timeout

48 Hours

Notification

Customer

## Chapter 7

# Transition Definition

A transition connects two states.



Open



Diagnosis

Each transition contains:

Source

Target

Condition

Permission

Actions

Events

## Chapter 8

# Conditions

Transitions occur only when conditions are satisfied.

Example

Estimated Cost



1000



Manager Approval

Another example

Parts Available

=

True

↓

Start Repair

Conditions are declarative.

## Chapter 9

### Actions

Actions execute automatically.

Examples

Assign Technician

Generate PDF

Create Invoice

Send Email

Send WhatsApp

Notify Customer

Generate AI Summary

Update Dashboard

Actions are reusable.

## Chapter 10

### Events

Every transition emits events.

Example

Repair Completed

↓

Warranty Generated

↓

Customer Notified

↓

Analytics Updated

↓

Bitey Learns

Events connect domains.

# Chapter 11

## Timers

Timers automate waiting periods.

Example

Diagnosis

↓

48 Hours

↓

Reminder

Example

Waiting Approval

↓

72 Hours

↓

Escalate

# Chapter 12

## Escalation

Escalation automatically changes responsibility.

Example

Waiting Technician

↓

24 Hours

↓

Supervisor

Example

Manager Approval

↓

48 Hours

↓

Director

## Chapter 13

### Parallel Workflows

Several branches may execute simultaneously.

Repair

├─ Order Parts

├─ Customer Notification

├─ Inventory Reservation

└─ Schedule Technician

Synchronization occurs automatically.

## Chapter 14

### Compensation

If one action fails, compensation restores consistency.

Example

Reserve Inventory

↓

Payment Failed

↓

Release Inventory

Compensation avoids inconsistent business states.

## Chapter 15

### Rollback

Rollback returns the workflow to a safe state.

Example

Repair Started



Critical Failure



Return to Diagnosis

Rollback is controlled by policy.

## Chapter 16

# Workflow Templates

The platform includes reusable templates.

Examples

Repair Ticket

Remote Support

On-site Service

Warranty Claim

Equipment Delivery

Sales Quote

Invoice Approval

Customer Complaint

Subscription Renewal

Templates accelerate deployment.

## Chapter 17

# AI Integration

Bitey participates in workflows as a decision support engine.

AI responsibilities include:

Intent Detection

Knowledge Retrieval

Workflow Recommendation

Risk Analysis

Customer Summary

Suggested Next Step

Automatic Classification

Priority Suggestion

Bitey recommends.

The workflow engine decides.



# Chapter 18

## SLA Management

Every workflow defines service-level objectives.

Example

| State            | Maximum Time |
|------------------|--------------|
| Open             | 2 Hours      |
| Diagnosis        | 24 Hours     |
| Waiting Approval | 48 Hours     |
| Repair           | 72 Hours     |
| Delivery         | 24 Hours     |

Violations generate events.

# Chapter 19

## Workflow Monitoring

Every execution records:

Workflow ID

Current State

Execution Time

Responsible User

Pending Actions

Events Generated

SLA Status

Execution History

Everything is auditable.

## Chapter 20

# Workflow Versioning

Workflows evolve independently.

Every version contains:

Workflow ID

Version

Effective Date

Author

Reviewer

Approval Date

Migration Notes

Running instances continue with their original version unless explicitly migrated.

# Chapter 21

## Workflow Governance

Every workflow change must be reviewed before publication.

Approval includes:

- Business validation.
- Technical validation.
- Security validation.
- AI impact assessment.
- Performance review.

No production workflow changes without governance.

# Chapter 22

## Long-Term Vision

The Workflow Engine should eventually allow business analysts to create or modify workflows through a graphical designer without changing code.

The execution engine will interpret these definitions at runtime.

## Architecture Principles

The Workflow Engine must remain:

- Stateless.
- Configurable.

- Event-driven.
- Domain-aware.
- Multi-tenant.
- Versioned.
- Auditable.
- AI-assisted.
- Technology-independent.

## Enterprise Vision

At maturity, BiteFixes should behave as a **Business Process Management Platform (BPM)** rather than a simple repair management application.

The Workflow Engine becomes the orchestration layer connecting:

- Customer Management.
- Ticket Management.
- Inventory.
- Finance.
- Notifications.
- Analytics.
- Integrations.
- Bitey AI.

This separation allows organizations to adapt their business processes without modifying application code, providing a level of flexibility typically found in enterprise-grade BPM suites.